

Title: Problem for Covid - 19 Data Analysis Project using Python

Dataset link :

Url = <https://raw.githubusercontent.com/SR1608/Datasets/main/covid-data.csv>

Perform following analysis on above dataset :

1. Import the dataset using Pandas from above mentioned url.

2. High Level Data Understanding:

- a. Find no. of rows & columns in the dataset
- b. Data types of columns.
- c. Info & describe of data in dataframe.

3. Low Level Data Understanding :

- a. Find count of unique values in location column.
 - b. Find which continent has maximum frequency using values counts.
 - c. Find maximum & mean value in 'total_cases'.
 - d. Find 25%,50% & 75% quartile value in 'total_deaths'.
 - e. Find which continent has maximum 'human_development_index'.
 - f. Find which continent has minimum 'gdp_per_capita'.
4. Filter the dataframe with only this columns ['continent','location','date','total_cases','total_deaths','gdp_per_capita','human_development_index'] and update the data frame.

5. Data Cleaning

- a. Remove all duplicates observations
 - b. Find missing values in all columns
 - c. Remove all observations where continent column value is missing
- Tip : using subset parameter in dropna
- d. Fill all missing values with 0

6. Date time format :

- a. Convert date column in datetime format using pandas.to_datetime

b. Create new column month after extracting month data from date column.

7. Data Aggregation:

a. Find max value in all columns using groupby function on 'continent' column

Tip: use reset_index() after applying groupby

**b. Store the result in a new dataframe named 'df_groupby'.
(Use df_groupby dataframe for all further analysis)**

8. Feature Engineering :

a. Create a new feature 'total_deaths_to_total_cases' by ratio of 'total_deaths' column to 'total_cases'

9. Data Visualization :

a. Perform Univariate analysis on 'gdp_per_capita' column by plotting

histogram using seaborn dist plot.

b. Plot a scatter plot of 'total_cases' & 'gdp_per_capita'

c. Plot Pairplot on df_groupby dataset.

d. Plot a bar plot of 'continent' column with 'total_cases' .

Tip : using kind='bar' in seaborn catplot

10. Save the df_groupby dataframe in your local drive using pandas.to_csv function .